

Nithin Kumar Reddy

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SUMMARY

- CS undergrad building **production-oriented AI systems**: multi-agent orchestration, cloud data pipelines (Azure/Databricks), and autonomous developer tooling (AST, LSP, RAG).
- Strong backend + systems focus — FastAPI, distributed ETL, real-time monitoring, and security-oriented open source (Prowler/AWS). Comfortable owning features end-to-end from design to demo.
- Looking for software engineering / platform / AI-infra internships where I can ship measurable reliability, automation, and developer-experience improvements.

EDUCATION

PES University

B.Tech, Computer Science and Engineering

Aug 2023 – May 2027

Bengaluru, Karnataka

Coursework: Data Structures & Algorithms, OOAD, DBMS, Operating Systems, Computer Networks, Compiler Design, Cloud Computing, Machine Learning, Big Data, Data Analysis

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, SQL, Bash

Backend & APIs: FastAPI, Node.js, Express, REST, WebSockets, JWT/OAuth, async services

Cloud & Data: Azure (ADF, Functions, ADLS Gen2), Databricks, Apache Spark, Kafka, AWS, Docker, Kubernetes, Linux, Git

AI/ML: Multi-agent systems, RAG, LangChain, Hugging Face Transformers, scikit-learn, TensorFlow, OpenCV, Pandas/NumPy

Frontend & Tooling: React, Playwright, FFmpeg, HTML/CSS

Databases: PostgreSQL, MySQL, MongoDB, SQLite, Firebase, Supabase

Enterprise: SAP S/4HANA Sales & Distribution (SD)

EXPERIENCE

Center for Information Security, Forensics and Cyber Resilience (IFSCR)

Jun 2025 – Aug 2025

Summer Research Intern — Digital Image Forensics

PES University

- Built an end-to-end image provenance pipeline that identifies the original source of manipulated/reposted images in a simulated social network (feature matching + temporal graph analysis).
- Modeled dissemination as a time-aware graph over users/posts and traced likely origin paths under reuploads, crops, and reposts.
- Implemented SIFT-based robust matching resilient to scale, crop, and rotation transforms; combined similarity + graph signals into a confidence-ranked source prediction.
- Delivered a research prototype with reproducible evaluation workflow; work informed forensic analysis methodology for multi-hop image reuse scenarios.

PROJECTS

Prowler (Cloud Security) — Open Source | Python, AWS, compliance checks

2025 – 2026

- External contributions to **Prowler** (widely used open-source cloud security platform) — not personal-only repositories.
- **PR #12076 (merged):** Configurable AWS EC2 check for instances left stopped beyond an admin-defined threshold.
- **PR #12118:** SageMaker endpoint-config **KMS encryption** check; service model + unit tests + changelog.

Memgraph — Open Source | C++, Cypher query planner

2026

- **PR #4490:** Fixed a SIGSEGV in the query planner for RETURN-only EXISTS subqueries (null input_op dereference on existential planning path).

OpenSRE (Tracer Cloud) — Open Source | Python, agent harness, testing

2026

- **PR #4401 (merged):** Unit tests for record_conversation_turn — append, window bounds, summary compaction, and in-place list updates.

RTK — Open Source | GitHub Actions, CI/CD

2026

- **PR #3251:** Guarded CD/release entry jobs so pre-release and release-please run only on the canonical repo, preventing fork pipeline failures.

Aegis Engineer | Python, multi-agent systems, AST, LSP, semantic search, GitHub API

2025 – 2026

- Building an autonomous software-engineering platform that ingests a repo, understands structure via AST/LSP + semantic search, and proposes production-ready PR-style fixes for GitHub issues.
- Designed multi-agent orchestration (repo intelligence, planning, code edit, verification) to reduce manual triage for issue resolution workflows.
- Focus: real developer tooling — not demo-only chat; emphasizes grounded repo context and reviewable diffs over free-form generation.
- github.com/Nithin078/Aegis-Engineer

Agentic Orchestration for Big Data Pipelines

Python, FastAPI, Azure Data Factory, Databricks, React, scikit-learn

Jan 2025 – May 2025

- Architected a **7-agent system** (Planner, Resource, Performance Prediction, Monitor, Assurance, Executor, Central Manager) to orchestrate ETL pipelines on Azure Data Factory + Databricks.
- Trained HistGradientBoosting models for automated compute sizing (workers, DIU, shuffle partitions), cutting over-provisioning by ~**30%** in evaluation runs.
- Shipped real-time monitoring (WebSockets), LLM-assisted anomaly detection (Groq), and mid-run reallocation; reduced simulated pipeline failure rate by ~**25%**.
- Built dual-validation Assurance Agent (structural checks + LLM semantic verification) so plans are checked against user intent before execution; React dashboard for cost/performance analytics via FastAPI.
- [GitHub repo](#)

DemoGen | Python, FastAPI, React, Playwright, FFmpeg, TTS

Mar 2026 – Apr 2026

- Built an AI platform that turns product walkthroughs into narrated demo videos: Playwright browser automation + screen capture + TTS + FFmpeg assembly.
- Engineered a reliable action runner for multi-step UI flows with status tracking and downloadable outputs through a FastAPI + React control plane.
- Useful for product/eng teams who need repeatable demos without manual recording each release.
- github.com/Nithin078/DemoGen

ACHIEVEMENTS & CERTIFICATIONS

Top 4 — FASTATHON, SRM University All-India Hackathon [[cert](#)]

Top 4 — Space Race Hackathon, Equinox Club, PES University

Participant — Adobe India Hackathon [[cert](#)]

Oracle Cloud Infrastructure — AI Foundations Associate [[badge](#)]

HackerRank — Problem Solving (Intermediate)